

Supporting Information

Auditing spatial-prior evaluation: coordinate frames, expression-only controls and specified interventions

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Author-review supplement, 1 October 2026. Scientific submission HOLD. This readable supplement exports committed records underlying the retained Results. It adds no tissue, method fit, model selection, resampling or independent validation. The machine-readable records retain full precision, source hashes, predictions and implementation identities. Historical broad benchmark outputs are not promoted into the corrected field panel.

S1. Retained coordinate identities and completed checks

The seven retained objects have explicit field or section identities, or a single registered spatial library. Table S1 records the keys used to keep neighbourhoods within a physical frame and the number of method–seed predictions checked against saved records. Raw-input checks reproduce selected cells, fields and labels. Zero cross-field links is an execution property, not evidence that the annotation is biologically correct. CODEX niches are cluster-derived proxies, not independent biological labels. Full source identities and selection hashes are in the protocol, summary and QA ledgers; three optimisation seeds are not three independent tissues.

Table S1: Field identities and committed source/prediction checks. Every object has seven procedures at seeds 1, 2 and 3. The single-library DLPFC input uses its registered spatial library rather than an observation column as its coordinate frame.

Object	Field key	Label key	Cross-field links	Rescored runs
DLPFC	uns:spatial	Region	0	21
MERFISH	slice_id	domain	0	21
STARmap	section_id	layer_label	0	21
MIBI	library_id	Cluster	0	21
Open-ST	section_id	ground_truth	0	21
CODEX	section_id	niche	0	21
Zhuang	section_id	domain	0	21

S2. Named-method contrasts and excluded counterfactual controls

Table S2 gives the three-seed mean of each named procedure’s raw ARI minus the genuinely expression-only PCA–KMeans ARI on the same cells and seed identities. These are fixed-protocol differences, not reference-label maxima across backends or refinement settings. The main article gives raw ARI and paired-seed SD separately. SD describes optimisation variation; no tissue-level interval is inferred from these seeds.

Table S2: Named raw-ARI differences from the expression-only control. “Mean” denotes neighbour-mean smoothing, “BANKSY” denotes the BANKSY-style implementation and “Leiden” denotes SpatialLeiden. CODEX Leiden returns 51/51/54 clusters against 100 reference classes; its entry is not an equal- K superiority comparison.

Object	Mean	BANKSY	Leiden	Tessera	STAGATE	SEDR
DLPFC	+0.063	+0.056	+0.275	+0.207	+0.060	+0.153
MERFISH	+0.130	+0.075	+0.011	+0.296	+0.066	+0.040
STARmap	+0.364	+0.331	+0.202	+0.336	+0.130	+0.212
MIBI	−0.058	−0.040	+0.020	−0.024	−0.033	−0.005
Open-ST	−0.047	−0.040	−0.003	−0.037	−0.027	−0.011
CODEX	−0.003	+0.003	+0.034	+0.004	+0.007	+0.000
Zhuang	+0.171	+0.134	+0.064	+0.104	+0.108	+0.074

The pooled-coordinate neighbour-mean arms and spatially refined nominal control are diagnostic counterfactuals, not eligible comparators. Neither enters these differences. SpaceFlow, GraphST and SpaGCN remain unavailable under the audited spatial path; no historical score is imputed. Frame provenance remains unestablished here for seqFISH, osmFISH, BRCA, IMC, Slide-seqV2, st_COAD, st_LIHC and st_OV. Their old outcomes, oracle maxima, proxy searches and component-removal analyses are outside current inference, not failed observations in this seven-object table.

S3. Row-order sensitivity from assigning ordinal ranks to ties

The unchanged historical eleven-object panel is a numerical counterexample only. For each of 200 row permutations, the joint partial association uses ranked covariates for class count, feature dimension and modality, an intercept, separate OLS rank residuals and their Pearson correlation. Assigning ordinal ranks to equal values makes this output row-order dependent; average midranks preserve ties (Table S3). The range is across computational row orders, not an uncertainty interval. Correct ranking does not repair the historical graph or contaminated control and does not establish a tissue-level mechanism.

Table S3: Joint partial-correlation range across 200 row permutations of one unchanged historical panel. Six-decimal printing collapses the midrank range, whose machine-level width is below 10^{-12} .

Rank operator	Minimum	Maximum
Ordinal tied ranks	0.584748	0.956280
Average midranks	0.928385	0.928385

S4. Coupled changes in the label-regeneration intervention

The endpoint audit recomputes generator quantities without fitting STAGATE. It reuses two substrates, five seed identities and intervention levels zero and one. Coordinates and the conditional expression-generation rule are fixed, while selected labels are redrawn and expression regenerated from the current labels. The operation changes class frequencies, contiguity and expression spatial autocorrelation together. Tables S4 and S5 report the substrate means and all 20 seed-level endpoint records, respectively. They are not additional independent tissues or independent-contiguity estimates.

Expression Moran’s I is averaged across features on the fixed directed six-neighbour graph. Residual SD is the population-form standard deviation of expression minus its generated label-conditioned mean. “Small class” is the final generator class, not a retrospectively selected minority. Random same-label probability is $\sum_c n_c(n_c - 1)/[n(n - 1)]$, where n_c is the generated count for class c and n is the substrate’s cell count. All four

readouts are dimensionless except residual SD, which retains the synthetic expression scale. Level is the label-redraw fraction, not measured expression–position alignment.

Table S4: Five-seed mean endpoints for label regeneration. The conditional rule remains fixed, while expression Moran’s I and class balance change.

Substrate	Level	Moran’s I	Small class (%)	Residual SD	Random same label
S1	0	0.062990	4.24	0.999165	0.185985
S1	1	−0.000068	16.16	1.004279	0.166535
S2	0	0.079996	4.31	0.999765	0.155621
S2	1	0.000083	13.94	1.001994	0.142980

Table S5: Seed-level endpoints underlying Table S4. Each seed contributes both levels; the rows are paired, not independent replications of a tissue mechanism.

Substrate	Seed	Level	Moran’s I	Small class (%)	Residual SD	Random same label
S1	1	0	0.057859	4.24	0.996725	0.185985
S1	1	1	−0.000729	16.00	1.003016	0.166797
S1	2	0	0.061848	4.24	0.999931	0.185985
S1	2	1	−0.001151	17.30	1.005459	0.166670
S1	3	0	0.067982	4.24	1.001084	0.185985
S1	3	1	0.002246	15.66	1.003055	0.166917
S1	4	0	0.064309	4.24	1.000426	0.185985
S1	4	1	−0.000284	15.57	1.006124	0.166123
S1	5	0	0.062950	4.24	0.997657	0.185985
S1	5	1	−0.000422	16.26	1.003742	0.166166
S2	1	0	0.065242	4.31	0.998583	0.155621
S2	1	1	−0.000191	14.31	1.001793	0.143106
S2	2	0	0.086628	4.31	0.998475	0.155621
S2	2	1	−0.001366	12.56	1.002805	0.142961
S2	3	0	0.087402	4.31	1.000952	0.155621
S2	3	1	−0.000028	15.31	1.001614	0.143159
S2	4	0	0.080750	4.31	1.002286	0.155621
S2	4	1	0.000995	13.31	1.003360	0.142770
S2	5	0	0.079960	4.31	0.998531	0.155621
S2	5	1	0.001007	14.19	1.000396	0.142906

Reproducibility and scientific boundary

The readable tables are exported by the current SI build script from ledgers/revision_20260923/ protocol, summary, QA and estimand-audit records. The export rejects incomplete objects, changed audit sources or missing endpoint identities and saves input and generated-table hashes. Those hashes establish identity, not scientific correctness. Independent review of the field-aware adapters, generator uncertainty and annotation scope remains open. No calibrated probability model is supplied for the returned KMeans labels; historical confidence and ECE claims stay withdrawn.